AI623 Final Project Proposal

Title: *Load-Bearing Steps in Flow Matching: Per-Class Analysis and Class-Aware Few-Step Schedules*

Group:7

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Problem Statement

Flow Matching (FM) models [Lipman et al., ICLR 2023] generate images by integrating a learned velocity field over many sampling steps. Reducing this to 1–4 steps is the central goal of recent few-step generation research, but current methods uniformly space sampling steps along the trajectory. **It is unknown which steps are actually load-bearing — i.e., which steps, if perturbed or skipped, materially change the final image — and whether load-bearing steps differ across image classes.** We hypothesize that (i) a small subset of steps dominates generation quality, (ii) this subset shifts across classes of differing visual complexity, and (iii) a class-aware non-uniform schedule allocating compute to load-bearing steps will outperform uniform spacing at matched step counts.

Approach

We train a class-conditional Flow Matching model with Classifier-Free Guidance (CFG) on CIFAR-10 from scratch using the OT-CFM objective (TorchCFM). We then run a *step ablation analysis*: at each sampling step *k*, we perturb the model's predicted velocity (zero-out, Gaussian noise, swap with a different class's prediction) and measure the resulting change in the final image (LPIPS, FID shift). This produces a per-class importance curve over sampling steps. We use these curves to design non-uniform 2-, 4-, and 8-step schedules (greedy and class-aware variants) and compare them against uniform spacing and DPM-style log-spacing baselines.

Pipeline (Block Diagram)

1. **Input:** CIFAR-10 dataset (32×32 images, 10 classes).
2. **Train** a class-conditional Flow Matching UNet with CFG on CIFAR-10 using TorchCFM → produces trained model θ.
3. **Step Ablation Analysis** on trained model θ:
    a. Sample images at 32 steps and log the full trajectories.
    b. At each step *k*, perturb the velocity prediction and measure ΔLPIPS and ΔFID on the final image.
    c. Aggregate results into per-class step-importance curves.
4. **Schedule Design & Evaluation:** Using the importance curves, construct Uniform, Log, Greedy, and Class-Aware schedules at 2, 4, and 8 NFEs.
5. **Output / Metrics:** Compare all schedules on FID, Precision/Recall, and per-class accuracy.

Dataset & Setting

CIFAR-10 (50K train, 10K test, 10 balanced classes). Small enough to train from scratch on a single GPU; widely used so results are comparable to prior work. Backbone: small UNet (~35M params) following the TorchCFM CIFAR-10 reference.

Evaluation

Primary metrics: **FID** (image quality), **Precision/Recall** [Kynkäänniemi et al. 2019] (fidelity vs. diversity), **per-class accuracy** of a pretrained CIFAR-10 classifier on generated images (prompt-following). Stress tests: (i) high-CFG regime ($w \in \{1, 3, 7\}$), (ii) hardest-class subset analysis, (iii) NFE-matched comparison against uniform/log/DPM schedules. **Failure analysis:** which classes degrade first under aggressive few-step sampling, and whether load-bearing curves correlate with intra-class visual variance.

Feasibility

Hardware: 1× RTX 5000 Ada (16 GB). FM training on CIFAR-10 with a small UNet takes ~24 GPU-hours to reach FID < 15 (per TorchCFM benchmarks). Step-ablation analysis is sampling-only (~12 GPU-hours). Total compute: <2 GPU-days, well within the 6-week window. No distillation, no large-scale text-to-image, no pretrained-checkpoint dependence — the project is fully self-contained on commodity hardware.

Why this is a Good Project

It asks one focused question, requires changing only one axis (the sampling schedule) over a clean baseline, includes both stress tests and per-class failure analysis, and produces an interpretable scientific result regardless of whether the proposed class-aware schedule beats uniform spacing.